

● RAG Learning

6.2 Averages, Spread & Grouped Data

AQA GCSE Mathematics 8300 — Statistics — spec refs S3, S4 — Papers 1 (non-calculator), 2 and 3 (calculator)

6.2.1 Types of Data (S4)

Primary data — collected by the person who is going to use it (own questionnaire, experiment, observation).

Secondary data — collected by someone else (census, websites, newspapers, databases).

Discrete data — can only take particular values, usually *counted* (number of goals, shoe size).

Continuous data — can take any value in a range, *measured* (height, mass, time).

	Advantages	Disadvantages
Primary	You know how it was collected; it answers exactly your question	Time-consuming and costly to collect
Secondary	Quick, cheap, often large data sets	May be out of date, biased or not quite what you need

6.2.2 Mean, Median, Mode and Range (S4)

$$\text{Mean} = \frac{\text{sum of values}}{\text{number of values}} \quad \text{Median} = \left(\frac{n+1}{2}\right) \text{th value (in order)}$$

Mode = most common value. **Range** = largest value — smallest value.

Worked Example: Data: 3, 8, 6, 10, 1, 4.

$$\text{Mean} = \frac{3 + 8 + 6 + 10 + 1 + 4}{6} = \frac{32}{6} = 5.33 \text{ (2 d.p.)}$$

In order: 1, 3, 4, 6, 8, 10. There are 6 values, so the median is the 3.5th value = $\frac{4 + 6}{2} = 5$.

There is no mode (every value occurs once). Range = $10 - 1 = 9$.

Problems using the mean

The key fact: $\text{total} = \text{mean} \times \text{number of values}$.

Worked Example: The mean of 6 numbers is 12. A seventh number is added and the mean becomes 13. Find the seventh number.

Original total = $6 \times 12 = 72$. New total = $7 \times 13 = 91$. Seventh number = $91 - 72 = 19$.

Worked Example: The mean height of 12 boys is 160 cm and of 8 girls is 150 cm. Mean of all 20:

$$\frac{12 \times 160 + 8 \times 150}{20} = \frac{1920 + 1200}{20} = \frac{3120}{20} = 156 \text{ cm. (Not 155 — you cannot just average the two means when the groups are different sizes.)}$$

Outliers and choosing an average

An **outlier** is a value much larger or smaller than the rest. It may be an error or a genuine but unusual value. Outliers distort the **mean** and the **range**; the median and IQR are hardly affected.

Average	Use when...	Drawback
Mean	Data is numerical with no outliers; uses every value	Distorted by outliers
Median	There are outliers (e.g. salaries, house prices)	Does not use all the values
Mode	Data is categorical, or the most common item is wanted (e.g. shoe sizes to stock)	There may be no mode or several modes

Worked Example: Salaries: £18 000, £19 500, £21 000, £22 000, £23 500, £95 000.

Mean = $199\,000 \div 6 = 33\,166.67$, i.e. £33 166.67. Median = $\frac{21\,000 + 22\,000}{2} = 21\,500$, i.e. £21 500.

The median is the better typical value: five of the six salaries are below the mean, which is pulled up by the outlier £95 000.

6.2.3 Averages from Frequency Tables (S4)

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

where x is each value and f is its frequency.

Worked Example: Number of pets owned by 20 people:

Pets x	Frequency f	$f \times x$	Running total of f
0	5	0	5
1	8	8	13
2	4	8	17

3	3	9	20
Total	20	25	

Mean = $25 \div 20 = 1.25$ pets. Mode = 1 (highest frequency).

Median: the 10.5th value. The running total shows values 6 to 13 are all 1, so the median = 1.

Range = $3 - 0 = 3$.

Exam Tip: A common error is dividing $\sum fx$ by the number of rows (4) instead of the total frequency (20). Another is giving the frequency (8) as the mode instead of the value (1 pet).

6.2.4 Grouped Frequency Tables (S4)

For grouped data the exact values are unknown, so:

- the **mean** is *estimated* using the **midpoint** of each class;
- the **modal class** is the class with the highest frequency;
- the **class containing the median** is found using a running total;
- the range can only be estimated (largest upper bound — smallest lower bound).

Worked Example: Lengths of 40 worms:

Length l (cm)	f	Midpoint x	fx	Running total
$0 < l \leq 5$	4	2.5	10	4
$5 < l \leq 10$	11	7.5	82.5	15
$10 < l \leq 15$	15	12.5	187.5	30
$15 < l \leq 20$	10	17.5	175	40

Total	40		455	
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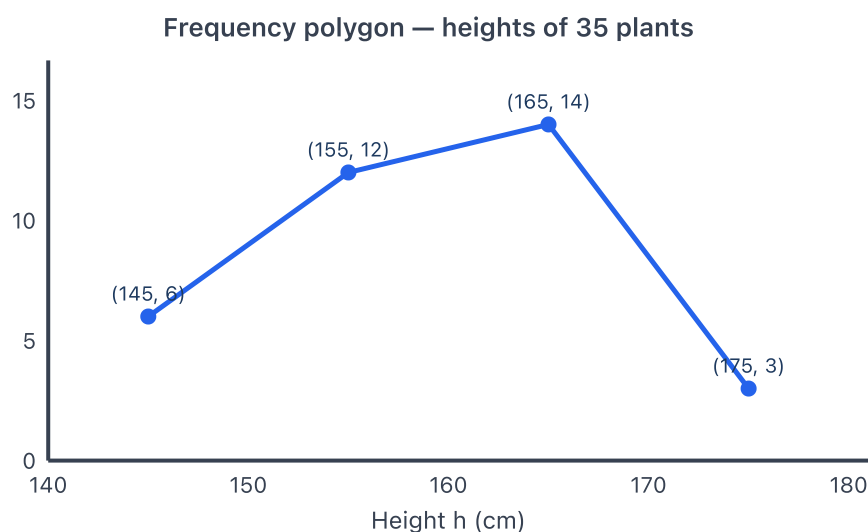
Estimated mean = $455 \div 40 = 11.375 \approx 11.4$ cm.

Modal class: $10 < l \leq 15$. Median: 20.5th value, which lies in $10 < l \leq 15$ (running totals 15 then 30).

Exam Tip: Check your estimated mean lies inside the range of the classes — here between 0 and 20. If you get 113.75, you have divided by the number of classes (4) instead of the total frequency (40).

Frequency polygons

A **frequency polygon** shows grouped data: plot each frequency at the **midpoint** of its class and join the points with straight lines. It is useful for comparing two distributions on the same axes.



6.2.5 Quartiles and Interquartile Range (S4)

Higher

Quartiles split ordered data into four equal parts.

- **Lower quartile (LQ)** — the $\left(\frac{n+1}{4}\right)$ th value; 25% of the data lies below it.
- **Median** — the $\left(\frac{n+1}{2}\right)$ th value.

- **Upper quartile (UQ)** — the $\left(\frac{3(n+1)}{4}\right)$ th value; 75% of the data lies below it.

$$\text{Interquartile range (IQR)} = \text{UQ} - \text{LQ}$$

The IQR measures the spread of the **middle 50%** of the data, so it is not affected by outliers.

Worked Example: Girls' times (s): 30, 34, 35, 39, 40, 42, 44, 46, 47, 50, 58 ($n = 11$)

LQ = 3rd value = 35. Median = 6th value = 42. UQ = 9th value = 47.

IQR = $47 - 35 = 12$ s. Range = $58 - 30 = 28$ s.

Worked Example: Data: 3, 5, 6, 8, 9, 11, 12, 14, 15, 18, 30. Is 30 an outlier if outliers are defined as more than $1.5 \times \text{IQR}$ above the UQ?

LQ = 6, UQ = 15, IQR = 9. $15 + 1.5 \times 9 = 28.5$. Since $30 > 28.5$, yes, 30 is an outlier.

6.2.6 Cumulative Frequency Graphs (S3) Higher

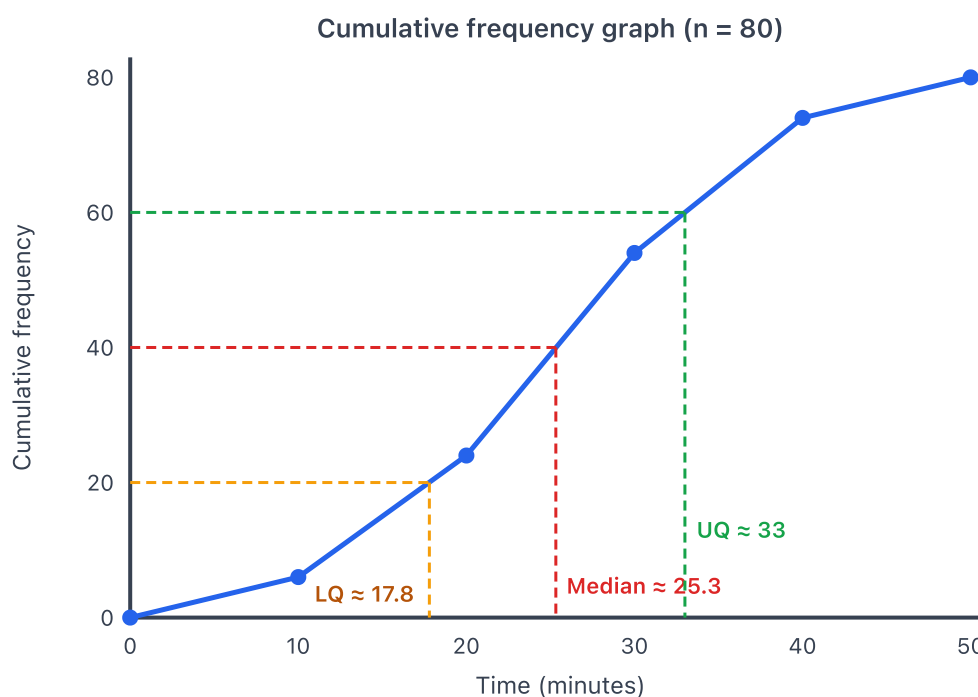
Cumulative frequency is the running total of the frequencies. A cumulative frequency graph plots each running total against the **upper class boundary**, starting at the lower boundary of the first class with cumulative frequency 0. Join the points with a smooth curve or straight line segments.

For total frequency n : read the **median** at $\frac{n}{2}$, the **LQ** at $\frac{n}{4}$ and the **UQ** at $\frac{3n}{4}$.

Worked Example: Times taken by 80 people to complete a puzzle:

Time t (min)	Frequency	Cumulative frequency	Plot at
$0 < t \leq 10$	6	6	(10, 6)
$10 < t \leq 20$	18	24	(20, 24)
$20 < t \leq 30$	30	54	(30, 54)
$30 < t \leq 40$	20	74	(40, 74)
$40 < t \leq 50$	6	80	(50, 80)

Also plot (0, 0).



From the graph: LQ ≈ 17.8 , median ≈ 25.3 , UQ ≈ 33 , so IQR $\approx 33 - 17.8 = 15.2$ minutes. (Readings from a hand-drawn curve will vary slightly; exam mark schemes accept a range.)

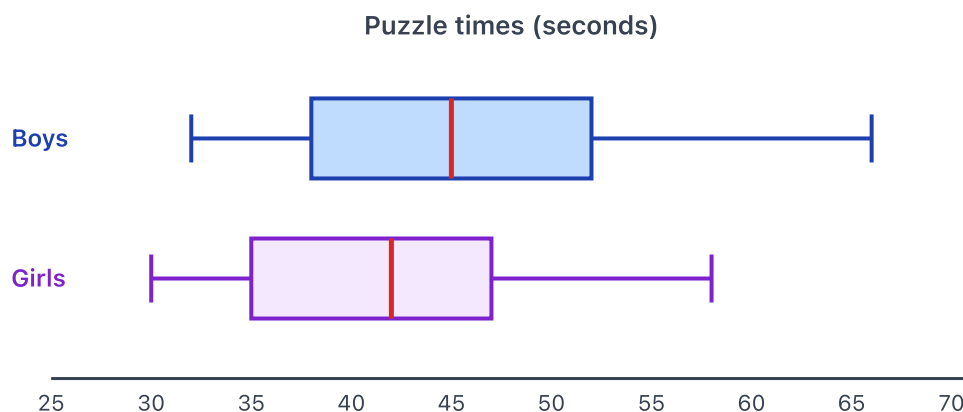
Worked Example: Reading "how many" questions. Estimate how many people took more than 35 minutes.

Read up from 35 minutes: cumulative frequency ≈ 64 . So $80 - 64 = 16$ people took more than 35 minutes.

Exam Tip: Plot at the **upper** class boundary, not the midpoint. Show your reading lines on the graph — method marks are often awarded for them. For "more than" questions, subtract your reading from the total.

6.2.7 Box Plots (S4) Higher

A **box plot** shows five values on a scale: **minimum, LQ, median, UQ, maximum**. The box runs from LQ to UQ (its length is the IQR) with a line at the median; whiskers extend to the minimum and maximum.



Worked Example: Boys: min 32, LQ 38, median 45, UQ 52, max 66. Girls: min 30, LQ 35, median 42, UQ 47, max 58.

Boys' IQR = $52 - 38 = 14$ s; girls' IQR = $47 - 35 = 12$ s.

Average: the girls' median (42 s) is lower than the boys' (45 s), so the girls were generally **faster**.

Spread: the girls' IQR (12 s) is smaller than the boys' (14 s), so the girls' times were **more consistent**.

Exam Tip: From a box plot you can say what fraction of the data lies in each part: 25% below the LQ, 50% between LQ and UQ, 75% below the UQ. You *cannot* read off the mean or the number of data values.

6.2.8 Histograms (S3) Higher

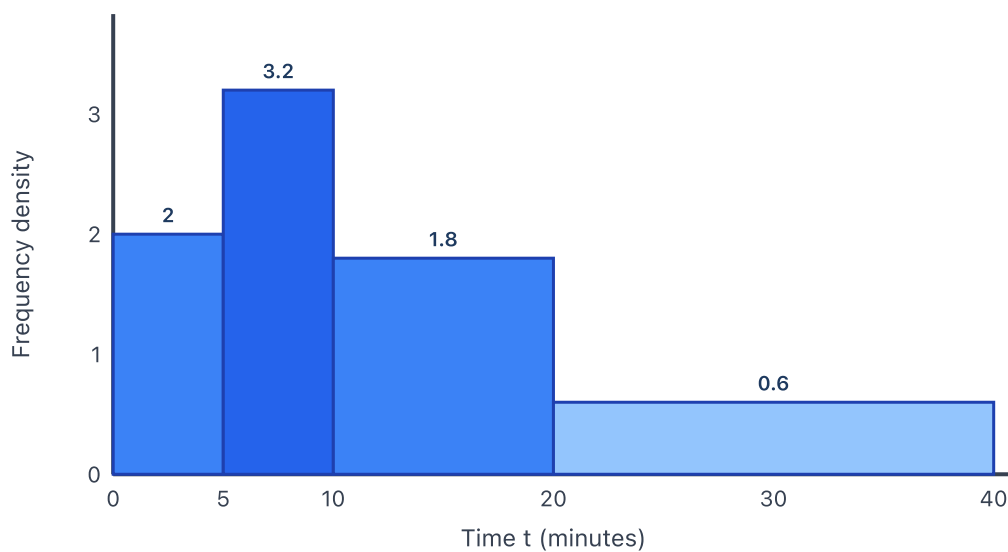
A **histogram** displays grouped continuous data. There are **no gaps** between bars, and the horizontal axis is a continuous scale. The **area** of each bar represents the frequency, so the vertical axis shows **frequency density**. With equal class widths the bar heights are proportional to frequency; with **unequal** class widths you must use frequency density.

$$\text{Frequency density} = \frac{\text{frequency}}{\text{class width}} \quad \text{Frequency} = \text{frequency density} \times \text{class width}$$

Worked Example: Times (minutes) for 56 tasks:

Time t (min)	Frequency	Class width	Frequency density
$0 < t \leq 5$	10	5	$10 \div 5 = 2$
$5 < t \leq 10$	16	5	$16 \div 5 = 3.2$
$10 < t \leq 20$	18	10	$18 \div 10 = 1.8$
$20 < t \leq 40$	12	20	$12 \div 20 = 0.6$

Histogram — task times



Worked Example: Estimating within part of a class. Estimate how many times are between 15 and 30 minutes.

15 to 20: width 5, FD 1.8 $\Rightarrow 5 \times 1.8 = 9$.

20 to 30: width 10, FD 0.6 $\Rightarrow 10 \times 0.6 = 6$.

Estimate = $9 + 6 = 15$ (assuming values are spread evenly within each class).

Worked Example: Finding the total from a histogram. Bars: 0–10 FD 1.2, 10–30 FD 2.5, 30–60 FD 0.8.

Frequencies: $10 \times 1.2 = 12$, $20 \times 2.5 = 50$, $30 \times 0.8 = 24$. Total = 86.

Exam Tip: If a histogram has no scale on the vertical axis, use a bar where you *know* the frequency to find the scale: e.g. if a bar of width 10 with height 4 squares has frequency 20, then each square of height represents FD 0.5. Label the axis "frequency density", never "frequency".

6.2.9 Comparing Distributions (S4)

To compare two data sets, make **two** comparisons, each with numbers and in context:

1. an **average** (mean or median) — which group is higher/lower on average;
2. a measure of **spread** (range, or IQR at Higher) — which group is more consistent (smaller spread).

Worked Example: Shop A's daily sales (£): 380, 400, 410, 420, 430, 440, 450. Shop B: median £385, range £310.

Shop A: median £420, range $450 - 380 = 70$, i.e. £70.

Shop A has the higher median (£420 vs £385), so its sales were higher on average.

Shop A has the smaller range (£70 vs £310), so its sales were more consistent.

Exam Tip: Use comparative words: "higher", "lower", "more consistent", "more varied". Say what it means in context (e.g. "the girls were faster" when lower times are better). Use the median and IQR when there are outliers, because the mean and range are distorted by extreme values.

Summary

Measure / diagram	Key fact
Mean	$\sum x \div n$; from a table $\sum fx \div \sum f$; grouped uses midpoints (estimate)
Median	$\left(\frac{n+1}{2}\right)$ th value in order
Mode / modal class	Most frequent value / class
Range	Largest — smallest; affected by outliers
Quartiles, IQR Higher	LQ $\left(\frac{n+1}{4}\right)$ th, UQ $\left(\frac{3(n+1)}{4}\right)$ th; IQR = UQ — LQ

Cumulative frequency Higher	Plot at upper class boundaries; read at $\frac{n}{4}, \frac{n}{2}, \frac{3n}{4}$
Box plot Higher	Min, LQ, median, UQ, max
Histogram Higher	FD = frequency $\div$ class width; area = frequency